

# Statistics

## Worksheet 1a (Non-Calculator)

### Mean, median and mode of ungrouped data and frequency table

#### 1. W20/qp11/q10

- a. A football team recorded the number of goals scored in each of their 20 games. The table shows the results.

|                        |   |   |   |   |   |
|------------------------|---|---|---|---|---|
| Number of goals scored | 0 | 1 | 2 | 3 | 4 |
| Frequency              | 6 | 5 | 4 | 4 | 1 |

- Write down the mode. [1]
  - Find the median. [1]
- b. In the football team
- the 2 tallest players have a mean mass of 75kg
  - the 8 shortest players have a mean mass of 60kg.

Calculate the mean mass of these 10 players. [2]

#### 2. S19/qp12/q3

The height of water, in cm, in a river is recorded every week for 10 weeks. The heights of the water, compared with its normal level, are listed below.

−45 −30 −35 0 5 −10 −20 40 20 25

- Work out the range. [1]
- Calculate the mean. [2]

### 3. W16/qp12/q12

A school recorded the number of absent students over a 50-day period. The results are given in the table.

| Number of absent students | 0  | 1  | 2 | 3 | 4 | 5 or more |
|---------------------------|----|----|---|---|---|-----------|
| Number of days            | 25 | 15 | 6 | 3 | 1 | 0         |

- Write down the mode. [1]
- Calculate the mean. [2]

### 4. W16/qp11/q13

During one day, the temperature, in  $^{\circ}\text{C}$ , was recorded every 2 hours. The twelve results are given below.

-3      -2      -1      1      2      4      5      4      2      0      -2      -2

For these results, find

- the median, [1]
- the mean, [1]
- the difference between the highest and the lowest of these temperatures. [1]

### 5. S16/qp11/q20

The number of goals scored in each of 50 football matches was recorded. The results are given in the table.

| Number of goals scored | 0  | 1  | 2 | 3 | 4 | 5 | 6 |
|------------------------|----|----|---|---|---|---|---|
| Frequency              | 16 | 11 | 9 | 7 | 6 | 0 | 1 |

For these results, find

- the mode, [1]
- the median, [1]
- the mean. [2]

6. S15/qp12/q12

Omar has a pack of number cards. He picks these five cards.



- a. Write down the mode of the five numbers. [1]
- b. He takes another card from the pack.
  - i. If the mean of the six numbers is -1, what number did he pick? [1]
  - ii. If the difference between the highest and lowest of the six numbers is 12, what are the two possible numbers he could have picked? [1]

7. W14/qp12/q3

In an experiment, a red die and a blue die were thrown 10 times. Each time, the score on the red die was subtracted from the score on the blue die. The results are given below.

5      -4      -3      4      0      2      -1      -3      3      -2

For these results, find

- a. the median, [1]
- b. the mean. [1]

## Answer Key

### 1. W20/qp11/q10

|           |    |   |                                           |
|-----------|----|---|-------------------------------------------|
| 10(a)(i)  | 0  | 1 |                                           |
| 10(a)(ii) | 1  | 1 |                                           |
| 10(b)     | 63 | 2 | <b>M1</b> for $2 \times 75 + 8 \times 60$ |

### 2. S19/qp12/q3

|      |    |   |                                                                                                                                                      |
|------|----|---|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3(a) | 85 | 1 |                                                                                                                                                      |
| 3(b) | -5 | 2 | <b>B1</b> for -50 seen<br><b>M1</b> for<br>$(-45 - 30 - 35 + 0 + 5 - 10 - 20 + 40 + 20 + 25) \div 10$ oe<br><br>If 0 scored, <b>SC1</b> for answer 5 |

### 3. W16/qp12/q12

|        |        |    |                                                                           |
|--------|--------|----|---------------------------------------------------------------------------|
| 12 (a) | 0      | 1  |                                                                           |
| (b)    | 0.8 oe | 2* | <b>M1</b> for $(15 \times 1 + 6 \times 2 + 3 \times 3 + 4 \times 1) / 50$ |

### 4. W16/qp11/q13

|        |                  |    |  |
|--------|------------------|----|--|
| 13 (a) | 0.5 oe           | 1  |  |
| (b)    | $\frac{2}{3}$ oe | 1* |  |
| (c)    | (-) 8            | 1  |  |

### 5. S16/qp11/q20

|        |        |    |                                                                                        |
|--------|--------|----|----------------------------------------------------------------------------------------|
| 20 (a) | 0      | 1  |                                                                                        |
| (b)    | 1      | 1  |                                                                                        |
| (c)    | 1.6 oe | 2* | <b>M1</b> for $(11 \times 1 + 9 \times 2 + 7 \times 3 + 6 \times 4 + 1 \times 6) / 50$ |

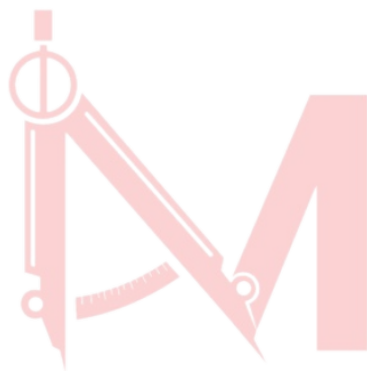
### 6. S15/qp12/q12

|         |       |   |              |
|---------|-------|---|--------------|
| 12 (a)  | -2    | 1 |              |
| (b) (i) | -3    | 1 |              |
| (ii)    | -8, 8 | 1 | Both correct |

7. W14/qp12/q3

$$\begin{array}{l} \textbf{3} \quad \textbf{(a)} \\ \textbf{(b)} \end{array} \left| \begin{array}{l} -0.5 \\ 0.1 \end{array} \right.$$

$$\left| \begin{array}{l} \textbf{1} \\ \textbf{1} \end{array} \right|$$



**Mathlete**  
= by Saad